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The leaf function of Penrose P2 graphs. (English) Zbl 08207624

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All graphs considered are simple and undirected. An induced subtree of a graph G is an induced subgraph that is a tree. The leaf function of G , denoted $L_G(n)$, is the maximum number of leaves over all induced subtrees of G on n vertices; an induced subtree attaining this maximum is called fully leafed. A Penrose P2-graph is the dual (adjacency) graph of a Penrose tiling by kites and darts: its vertices are the tiles, and two tiles are adjacent whenever they share a side. *C. Porrier* and *A. Blondin Massé* [“The Leaf function of graphs associated with Penrose tilings”, *Int. J. Graph Comput.* 1, No. 1, 1–24 (2020; doi: 10.35708/GC1868-126721)] showed that $2\varphi n/(4\varphi + 1) \leq L_{P2}(n) \leq \lfloor n/2 \rfloor + 1$, where φ is the golden ratio. In this paper, the authors determine the leaf function L_{P2} exactly, showing that

$$L_{P2}(n) = \begin{cases} 0 & \text{if } n \in \{0, 1\}, \\ \lfloor n/2 \rfloor + 1 & \text{if } 2 \leq n \leq 18, \\ L_{P2}(n - 17) + 8 & \text{if } n \geq 19. \end{cases}$$

In particular $L_{P2}(n) \sim 8n/17$. Since a Penrose tiling is aperiodic, it is perhaps surprising that L_{P2} satisfies a linear recurrence at all; the authors remark that they had expected the answer to depend on the irrationality of φ .

The upper bound builds on the result of Porrier and Blondin Massé [loc. cit.] that $\Delta(I) \leq 3$ for any induced subtree I of a P2-graph. The authors then prove that any 3-internal-regular induced subtree of a P2-graph has at most eight internal vertices and, in particular, is a caterpillar. A minimal counterexample, together with a factorization or graft operation that splits I along an edge into two induced subtrees meeting at that edge, yields the recurrence.

The lower bound is obtained by exhibiting a fully leafed caterpillar of every order. The base object is a caterpillar C_{116} on 116 vertices, assembled from seven copies of a caterpillar C_{14} , joined by three additional vertices at each of six junctions. That C_{116} occurs in a P2 tiling at all, and that the construction iterates, is established through the substitution structure: C_{116} arises as a φ^3 -decomposition of C_{14} , and the local isomorphism property of Penrose tilings then guarantees that any patch occurring once occurs in every P2 tiling. The authors organize this by passing to the associated HBS (hexagon-boat-star) tilings, decorating their vertices according to the type of the surrounding patch, so that the caterpillars are read off as paths in this coarser tiling. The authors also remark that there exist fully leafed induced subtrees that are not caterpillars.

The authors conclude with two conjectures to the effect that the fully leafed induced subtrees are essentially sub-caterpillars of the extremal family constructed using C_{116} . The authors also note that the leaf function L_{P3} of the Penrose P3 tiling differs from L_{P2} already at $n = 12$, and ask whether fully leafed induced subtrees of the two types can be transformed into one another. Determining the leaf function for the remaining tilings of the Penrose MLD class, and for substitutive tilings generally, remains open.

The closed formula given in Proposition 3 is incorrect for $n \equiv 0, 1 \pmod{17}$ with $n \geq 36$, the smallest instances being $n = 51$ and $n = 52$, where it returns 24 in place of the correct values 25 and 26. The error occurs in the last display of the proof: writing $n = 17q + r$ with $r = n \bmod 17$, the step $L_{P2}(r + 17) = L_{P2}(r) + 8$ requires $r + 17 \geq 19$, and so fails for $r \in \{0, 1\}$, where in fact $L_{P2}(17) = 9$ and $L_{P2}(18) = 10$. The corrected values are $8q + 1$ and $8q + 2$, respectively. Shifting the residue to the range $2 \leq r \leq 18$ repairs the statement and at the same time removes the need for a separate case, giving:

$$L_{P2}(n) = \begin{cases} 0 & \text{if } n \in \{0, 1\}, \\ 8\lfloor (n - 2)/17 \rfloor + \lfloor ((n - 2) \bmod 17)/2 \rfloor + 2 & \text{if } n \geq 2. \end{cases}$$

Some other minor typos are listed below. In Figure 4, the dark outline in the second kingdom does not correctly mark the deuce. In the algorithm of Section 4.4, the counter k is incremented at every iteration,

including those in which the pointer merely advances along the derived path and no vertex is added, so that the output has fewer than n vertices; the loop should instead be controlled by the order of T . Read “The leaf function of the P2-graph” in the statement of Corollary 2. In the proof of Proposition 3, the first displayed equation should read

$$L_{P2}(n) = \lfloor (n-17)/2 \rfloor + 1 + 8 = \lfloor (n+1-18)/2 \rfloor + 9 = \lfloor (n+1)/2 \rfloor.$$

The graded poset S appearing in Conjecture 2(i) is not defined.

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MSC:

- [05C05](#) Trees
- [05C35](#) Extremal problems in graph theory
- [52C23](#) Quasicrystals and aperiodic tilings in discrete geometry
- [52C20](#) Tilings in 2 dimensions (aspects of discrete geometry)
- [05C30](#) Enumeration in graph theory
- [05C63](#) Infinite graphs

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[Penrose](#); [kite and dart](#); [fully leafed induced subtrees](#); [tiling](#); [poset](#)

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References:

- [1] H. Abbott and M. Katchalski. On the snake in the box problem. *Journal of Combinatorial Theory, Series B*, 45(1):13-24, 1988. · [Zbl 0662.05034](#)
- [2] M. Baake. A guide to mathematical quasicrystals, 1999. URL <https://arxiv.org/abs/math-ph/9901014>.
- [3] M. Baake and U. Grimm. *Aperiodic Order. Vol 1. A Mathematical Invitation*, volume 149 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 9 2013. URL <http://oro.open.ac.uk/38933/>. · [Zbl 1295.37001](#)
- [4] A. Blondin Massé, J. de Carufel, A. Goupil, M. Lapointe, É. Nadeau, and É. Vandomme. Fully leafed induced subtrees. In C. Iliopoulos, H. W. Leong, and W.-K. Sung, editors, *Combinatorial Algorithms*, pages 90-101, Cham, 2018a. Springer International Publishing. ISBN 978-3-319-94667-2. doi: 10.1007/978-3-319-94667-2_8. · [Zbl 1515.05090](#) · doi:10.1007/978-3-319-94667-2_8
- [5] A. Blondin Massé, J. de Carufel, A. Goupil, and M. Samson. Fully leafed tree-like polyominoes and polycubes. In L. Brankovic, J. Ryan, and W. F. Smyth, editors, *Combinatorial Algorithms*, pages 206-218, Cham, 2018b. Springer International Publishing. ISBN 978-3-319-78825-8. · [Zbl 1507.05021](#)
- [6] A. Blondin Massé, J. de Carufel, and A. Goupil. Saturated fully leafed tree-like polyforms and poly-cubes. *Journal of Discrete Algorithms*, 52-53:38-54, 2018. ISSN 1570-8667. doi: <https://doi.org/10.1016/j.jda.2018.11.004>. URL <https://www.sciencedirect.com/science/article/pii/S1570866718301515>. *Combinatorial Algorithms -Special Issue Devoted to Life and Work of Mirka Miller*. · [Zbl 1403.05027](#) · doi:10.1016/j.jda.2018.11.004
- [7] O. Bodroza-Pantic, B. Pantic, I. Pantic, and M. Bodroza-Solarov. Enumeration of hamiltonian cycles in some grid graphs. *MATCH Commun. Math. Comput. Chem.*, (70):181-204, 2013. · [Zbl 1299.05203](#)
- [8] H. S. Chan and K. A. Dill. Compact polymers. *Macromolecules*, 22(12):4559-4573, 1989. URL <https://doi.org/10.1021/ma00202a031>. · doi:10.1021/ma00202a031
- [9] A. Chinnasamy, B. Sivakumar, P. Selvakumari, and A. Suresh. Minimum connected dominating set based rsu allocation for smartcloud vehicles in vanet. *Cluster Computing*, 22:12795-12804, 2019.
- [10] L. C. Collins, T. G. Witte, R. Silverman, D. B. Green, and K. K. Gomes. Imaging quasiperiodic electronic states in a synthetic penrose tiling. *Nat Commun*, 8(15961), 2017. URL <https://doi.org/10.1038/ncomms15961>. · doi:10.1038/ncomms15961
- [11] N. G. de Bruijn. Algebraic theory of penrose’s non-periodic tilings of the plane. i. *Indagationes Mathematicae (Proceedings)*, 84(1):39-52, 1981. ISSN 1385-7258. doi: [https://doi.org/10.1016/1385-7258\(81\)90016-0](https://doi.org/10.1016/1385-7258(81)90016-0). URL <https://www.sciencedirect.com/science/article/pii/S1385725881900160>. · doi:10.1016/1385-7258
- [12] R. D. Diehl, W. Setyawan, and S. Curtarolo. Gas adsorption on quasicrystalline surfaces. *Journal of Physics: Condensed Matter*, 20(31):314007, jul 2008. URL <https://dx.doi.org/10.1088/0953-8984/20/31/314007>. · doi:10.1088/0953-8984/20/31/314007
- [13] J.-M. Dubois. New prospects from potential applications of quasicrystalline materials. *Materials Science and Engineering: A*, 294-296:4-9, 2000. ISSN 0921-5093. doi: [https://doi.org/10.1016/S0921-5093\(00\)01305-8](https://doi.org/10.1016/S0921-5093(00)01305-8). URL <https://www.sciencedirect.com/science/article/pii/S0921509300013058>. · doi:10.1016/S0921-5093
- [14] L. M. Fernandes and L. Gouveia. Minimal spanning trees with a constraint on the number of leaves. *European Journal of Operational Research*, 104(1):250-261, 1998. · [Zbl 0957.90010](#)
- [15] F. Gähler, M. Baake, and M. Schlottmann. Binary tiling quasicrystals and matching rules. *Phys. Rev. B*, 50:12458-12467,

- Nov 1994. URL <https://doi.org/10.1103/PhysRevB.50.12458>. · [Zbl 0909.52006](#) · [doi:10.1103/PhysRevB.50.12458](#)
- [16] M. Gardner. Mathematical games. *Scientific American*, 236(1):110-121, 1977. ISSN 00368733, 19467087.
- [17] T. Garel and H. Orland. Mean-field model for protein folding. *Europhysics Letters*, 6(4):307, jun 1988. URL <https://dx.doi.org/10.1209/0295-5075/6/4/005>. · [doi:10.1209/0295-5075/6/4/005](#)
- [18] B. Grünbaum and G. C. Shephard. *Tilings and Patterns*. Dover Publications, second edition, 2016.
- [19] J. J. Kweon, H.-I. Kim, S. hwa Lee, J. Kim, and S. K. Lee. Quantitative probing of hydrogen environments in quasicrystals by high-resolution nmr spectroscopy. *Acta Materialia*, 226:117657, 2022. ISSN 1359-6454. doi: <https://doi.org/10.1016/j.actamat.2022.117657>. URL <https://www.sciencedirect.com/science/article/pii/S1359645422000428>. · [doi:10.1016/j.actamat.2022.117657](#)
- [20] H. Li, R. Helling, C. Tang, and N. Wingreen. Emergence of preferred structures in a simple model of protein folding. *Science*, 273(5275):666-669, 1996. doi: 10.1126/science.273.5275.666. URL <https://www.science.org/doi/abs/10.1126/science.273.5275.666>. · [doi:10.1126/science.273.5275.666](#)
- [21] R. McGrath, J. A. Smerdon, H. R. Sharma, W. Theis, and J. Ledieu. The surface science of quasicrystals. *Journal of Physics: Condensed Matter*, 22(8):084022, feb 2010. URL <https://dx.doi.org/10.1088/0953-8984/22/8/084022>. · [doi:10.1088/0953-8984/22/8/084022](#)
- [22] C. Porrier. HBS Tilings Extended: State of the Art and Novel Observations. *Electronic Proceedings in Theoretical Computer Science*, 403:156-163, June 2024. ISSN 2075-2180. URL <http://dx.doi.org/10.4204/EPTCS.403.32>. · [Zbl 1571.52020](#) · [doi:10.4204/EPTCS.403.32](#)
- [23] C. Porrier and A. Blondin Massé. The leaf function for graphs associated with penrose tilings. In *2019 First International Conference on Graph Computing (GC)*, pages 37-44, 2019. doi: 10.1109/GC46384.2019.00014. · [doi:10.1109/GC46384.2019.00014](#)
- [24] C. Porrier and A. Blondin Massé. The Leaf function of graphs associated with Pen-rose tilings. *International Journal of Graph Computing*, 1.1:1-24, 2020. doi: 10.35708/GC1868-126721. URL https://www.kspress.org/_files/ugd/e49175_. · [doi:10.35708/GC1868-126721](#)
- [25] E. A. Robinson. Symbolic dynamics and tilings of $\mathbb{R}^d$. *Proceedings of Symposia in Applied Mathematics*, 60, 01 2004. doi: 10.1090/psapm/060/2078847. · [doi:10.1090/psapm/060/2078847](#)
- [26] M. Senechal. *Quasicrystals and Geometry*. Cambridge University Press, 1995. · [Zbl 0828.52007](#)
- [27] A. Shioura, A. Tamura, and T. Uno. An optimal algorithm for scanning all spanning trees of undirected graphs. *SIAM Journal on Computing*, 26(3):678-692, 1997. doi: 10.1137/S0097539794270881. URL <https://doi.org/10.1137/S0097539794270881>. · [Zbl 0870.05066](#) · [doi:10.1137/S0097539794270881](#)
- [28] T. N. D. Silva and V. Sivised. A statistical mechanics perspective for protein folding from q-state potts model, 2018. URL <https://arxiv.org/abs/1709.04813>.
- [29] E. Vedmedenko, S. E.-D. Mandel, and R. Lifshitz. In search of multipolar order on the penrose tiling. *Philosophical Magazine*, 88(13-15):2197-2207, 2008. URL <https://doi.org/10.1080/14786430802302059>. · [doi:10.1080/14786430802302059](#)

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